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HSSC-I NATIONAL BOOK FOUNDATION FBISE

CHEMISTRY TEST - CHAPTER 11: NITROGEN AND SULPHUR

Name:		Roll No.		Section:

Total Time: 2:30 Hour
Time Allowed: 150 Minutes

Total Marks: 150

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. Nitrogen belongs to which group of the periodic table?	Group 17	Group 13	Group 15 (V)	Group 16	○○●○
2. Nitrogen makes up about what percentage of the Earth's atmosphere?	21%	1%	78%	0.04%	○○●○
3. Nitrogen exists in nature as:	monatomic atoms	diatomic molecules (N_2)	polyatomic rings (N_8)	triatomic molecules	○●○○
4. Nitrogen gas is chemically unreactive because of its:	very strong triple bond	low molecular mass	presence of lone pairs only	ionic character	●○○○
5. Nitrogen atoms need to gain how many electrons to achieve the noble gas configuration of neon?	3	1	2	5	●○○○
6. In the nitrogen molecule, the two atoms are held by a:	single covalent bond	ionic bond	triple covalent bond	double covalent bond	○○●○
7. The bond energy of the N≡N bond is approximately:	10 kJ mol^{-1}	1000 kJ mol^{-1}	100 kJ mol^{-1}	10 000 kJ mol^{-1}	○●○○
8. Nitrogen gas shows inert behaviour because the N_2 molecule is also:	ionic	highly polar	non-polar	metallic	○○●○
9. Nitrogen gas reacts only under:	extreme conditions	dilute acids	ordinary room conditions	moist air only	●○○○
10. Ammonia gas turns moist red litmus paper:	green	red	colourless	blue	○○○●
11. Ammonia is basic in nature because it has:	a lone pair on the nitrogen atom	a positive charge	an incomplete octet	no hydrogen atoms	●○○○
12. In the Brønsted-Lowry theory, ammonia is a base because it:	accepts an electron pair only	donates a neutron	accepts a proton	donates a proton	○○●○
13. In the Lewis concept, ammonia is a base because it is:	a neutron acceptor	a proton donor	an electron pair donor	an electron pair acceptor	○○●○
14. The ammonium ion is formed when ammonia reacts with:	an alkali metal	a base	a dilute acid	a noble gas	○○●○

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Questions	A	B	C	D	A B C D
15. In the formation of the NH_4^+ ion, the fourth N–H bond is a:	metallic bond	coordinate covalent bond	ionic bond	hydrogen bond	○●○○
16. The shape of the ammonium ion is:	trigonal pyramidal	tetrahedral	linear	planar	○●○○
17. In the laboratory, ammonia is prepared by heating an ammonium salt with a:	metal	base	noble gas	dilute acid	○●○○
18. Which pair of solids is heated to prepare ammonia in the laboratory?	NH_4Cl and HCl	NH_4Cl and $Ca(OH)_2$	NaOH and KOH	NaCl and $Ca(OH)_2$	○●○○
19. The balanced equation for the laboratory preparation of ammonia is:	$2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2NH_3 + 2H_2O$	$N_2 + 3H_2 \rightarrow 2NH_3$	$NH_4Cl + NaOH \rightarrow NaCl + NH_3$	$NH_3 + HCl \rightarrow NH_4Cl$	●○○○
20. In the laboratory preparation of ammonia, the gas formed is dried over:	quicklime (CaO)	P_4O_{10}	anhydrous $CaCl_2$	concentrated H_2SO_4	●○○○
21. Concentrated sulphuric acid is not suitable for drying ammonia because ammonia:	is acidic	dissolves in it without reaction	reacts with it to form ammonium sulphate	is oxidised by it to nitrogen	○○●○
22. Industrially, ammonia is manufactured by the:	Ostwald process only	Solvay process	Contact process	Haber process	○○○●
23. In the Haber process, the reactants are:	hydrogen and chlorine	nitrogen and hydrogen	nitrogen and oxygen	ammonia and oxygen	○●○○
24. The Haber process is a reversible reaction that is:	endothermic	neutral	exothermic	photochemical	○○●○
25. In the Haber process, a low temperature gives a higher yield but:	ammonia decomposes immediately	the reaction rate becomes slow	the pressure drops	the catalyst melts	○●○○
26. The temperature used in the Haber process (as stated in the book) is:	1000 °C	25 °C	500 °C	100 °C	○○●○
27. The pressure used in the Haber process (as stated in the book) is:	1 atm	200 atm	2 atm	10 atm	○●○○
28. High pressure favours the Haber process because the forward reaction:	has no change in moles	increases the number of moles of gas	is endothermic	decreases the number of moles of gas	○○○●
29. The catalyst used in the Haber process is:	nickel	platinum	vanadium(V) oxide	iron	○○○●
30. Hydrogen for the Haber process is usually obtained from methane and:	steam	nitrogen	oxygen	chlorine	●○○○
31. Nitrogen monoxide is formed naturally during:	snowfall	thunderstorms	volcanic cooling only	earthquakes only	○●○○

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Questions	A	B	C	D	A B C D
32. Lightning provides the ___ needed for nitrogen and oxygen to react.	activation energy	pressure	catalyst	solvent	●○○○
33. In internal combustion engines, nitrogen monoxide is formed when nitrogen reacts with oxygen at:	zero pressure	high temperature	low temperature	room temperature	○●○○
34. Nitrogen monoxide is oxidised by oxygen in the air to give:	nitrogen dioxide	nitric acid directly	nitrous oxide	ammonia	●○○○
35. Nitrogen dioxide dissolves in rain water to form:	sulphuric acid	ammonia	nitric acid	nitrous oxide	○○●○
36. Oxides of nitrogen released from vehicles contribute to:	global cooling	noble gas pollution	ozone formation in soil	acid rain	○○○●
37. Nitric acid in rain falls to earth and forms nitrates, which are used by:	animals to make carbon	plants to make proteins	rocks to form minerals	fish to produce hydrogen	○●○○
38. Nitrogen monoxide is removed from exhaust gases by a:	three-way catalytic converter	fractionating column	cooling tower	distillation tower	●○○○
39. The catalyst of a three-way catalytic converter is typically composed of:	platinum, palladium and rhodium	sodium, potassium and lithium	copper, zinc and tin	iron, nickel and cobalt	●○○○
40. In the catalytic converter, NO reacts with CO to form:	NH_3 and CO_2	NO_2 and C	N_2O and CO	N_2 and CO_2	○○○●
41. A balanced equation for the reaction in the catalytic converter between CO and NO is:	$CO + NO \rightarrow N_2 + CO$	$2CO + 2NO \rightarrow N_2 + 2CO_2$	$CO + 2NO \rightarrow N_2 + CO_3$	$2CO + NO \rightarrow N_2O + C$	○●○○
42. The three gases removed by a three-way catalytic converter are CO, hydrocarbons and:	sulphur dioxide	oxygen	carbon dioxide	nitrogen oxides	○○○●
43. The catalytic removal of NO produces nitrogen, which is:	a greenhouse gas	a harmless component of the atmosphere	an acidic gas	a toxic gas	○●○○
44. Carbon monoxide in exhaust gases is hazardous because it:	causes acid rain	combines with haemoglobin and reduces oxygen transport	forms ozone	damages the ozone layer	○●○○
45. Photochemical smog is formed from oxides of nitrogen and volatile organic compounds in the presence of:	darkness	cold air	rain	sunlight	○○○●
46. Photochemical smog appears as a:	blue fog	brown haze	clear sky	white mist	○●○○
47. Photochemical smog occurs more often in:	monsoon only	winter	spring only	summer	○○○●
48. Smog is a mixture of:	smoke and steam	fog and rain	smoke and fog	dust and wind	○○●○

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Questions	A	B	C	D	A B C D
49. PAN stands for:	peroxyacetyl nitrate	phenyl acetyl nitrate	peroxyacid nitrite	polyacetyl nitrate	●○○○
50. PAN is a ___ pollutant.	primary	natural only	secondary	radioactive	○○●○
51. PAN is an irritant to the:	liver only	bones	eyes and respiratory system	skin only	○○●○
52. In photochemical smog formation, NO_2 undergoes photodissociation into NO and:	an N atom	CO	an oxygen atom	a hydrogen atom	○○●○
53. The oxygen atom released from NO_2 reacts with unburned hydrocarbons to form:	nitrogen gas	sulphur dioxide	peroxyacyl radicals	ammonia	○○●○
54. Peroxyacyl radicals react with NO_2 to form:	ammonia	nitric acid	ozone only	PAN	○○○●
55. PAN can be transported over long distances because it is:	a solid	a metal	a stable compound	a very unstable gas	○○●○
56. In the morning hours, the levels of NO and hydrocarbons:	remain zero	increase, followed by an increase in NO_2	change only at night	decrease sharply	○●○○
57. Nitrification is an ___ process.	reduction	oxidation	combustion	neutralisation	○●○○
58. In nitrification, ammonium ions are first oxidised to:	nitrite ions	nitrogen gas	nitrous oxide	nitrate ions	●○○○
59. In nitrification, nitrite ions are further oxidised to:	nitrate ions	ammonia	ammonium ions	nitrogen gas	●○○○
60. The bacteria which convert ammonia into nitrites in nitrification are:	Clostridium	E. coli	Pseudomonas	Nitrosomonas	○○○●
61. The conversion of nitrites into nitrates is carried out by:	Pseudomonas	Clostridium	Nitrobacter	Nitrosomonas	○○●○
62. In nitrification, the oxidation state of nitrogen changes from:	−3 to +3 and then to +5	+5 to +3 and then to −3	0 to +3	+3 to 0	●○○○
63. Denitrification is a biological process in which:	ammonia is oxidised to nitrite	nitrogen gas is converted to ammonia	nitrate is reduced to nitrogen gas	ammonia is converted to nitrate	○○●○
64. Denitrification is carried out in the ___ of oxygen.	large amounts	absence	excess	presence	○●○○
65. Which bacteria take part in denitrification?	E. coli and Salmonella	Lactobacillus and Rhizobium	Nitrosomonas and Nitrobacter	Clostridium and Pseudomonas	○○○●

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Questions	A	B	C	D	A B C D
66. Denitrification is the ___ stage of the nitrogen cycle.	middle	final	first	only	○●○○
67. Nitrification is the ___ stage of the nitrogen cycle.	middle only	last	unimportant	first and essential	○○○●
68. Plants absorb nitrogen from the soil mainly in the form of:	ammonia gas	nitrogen gas	nitrous oxide	nitrates	○○○●
69. Which oxide of nitrogen is released from car engines?	Nitrogen pentoxide	Nitrous oxide	Nitric oxide	Nitrogen tetroxide	○○●○
70. Sulphur lies below oxygen in Group:	16 (VI)	17 (VII)	15 (V)	14 (IV)	●○○○
71. The molecular formula of elemental sulphur is:	S_8	S	S_4	S_2	●○○○
72. Sulphur molecules are:	triatomic bent	cyclic octatomic	tetrahedral	linear diatomic	○●○○
73. Elemental sulphur at room temperature is a:	black powder	colourless gas	white liquid	bright yellow crystalline solid	○○○●
74. Sulphur has how many electrons in its valence shell?	2	4	6	8	○○●○
75. Possible oxidation states of sulphur are:	−3, +3 and +5	−2, +2, +4 and +6	−1, 0 and +1	+1 and +3 only	○●○○
76. The most stable oxidation state of sulphur given in the book is:	−2	+2	+6	+4	○○●○
77. Sulphur in sulphuric acid has an oxidation state of:	+6	+2	−2	+4	●○○○
78. Sulphur in sulphur dioxide has an oxidation state of:	+4	−2	+6	+2	●○○○
79. Sulphur in hydrogen sulphide has an oxidation state of:	+4	0	+2	−2	○○○●
80. Elemental sulphur is relatively unreactive because:	it has a very low boiling point	it has no electrons in the valence shell	it is a gas	a lot of energy is needed to break the S–S bonds in S_8	○○○●
81. When sulphur gains two electrons, it acts as:	an oxidising agent	a catalyst	a reducing agent	an acid	●○○○
82. When sulphur loses electrons to reach +4 or +6, it acts as:	a reducing agent	an oxidising agent	a base	a catalyst	●○○○

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Questions	A	B	C	D	A B C D
83. In the reaction $S + H_2 \rightarrow H_2S$, sulphur acts as:	a catalyst	a reducing agent	an oxidising agent	a solvent	○○●○
84. In the reaction of sulphur with zinc or iron, the oxidation state of sulphur changes from:	0 to +2	+4 to +6	−2 to 0	0 to −2	○○○●
85. Sulphur burns in air with a:	red flame	green flame	blue flame	yellow flame	○○●○
86. Sulphur burning in air forms:	hydrogen sulphide	sulphuric acid	sulphur trioxide	sulphur dioxide	○○○●
87. In the reaction $S + O_2 \rightarrow SO_2$, sulphur acts as:	a catalyst	a base	a reducing agent	an oxidising agent	○○●○
88. In the reaction $S + 6HNO_3 \rightarrow H_2SO_4 + 6NO_2 + 2H_2O$, sulphur is oxidised from:	0 to +6	0 to +4	+4 to +6	−2 to +6	●○○○
89. In the reaction $S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$, sulphur is oxidised from:	0 to +4	−2 to +4	+6 to +4	0 to +6	●○○○
90. Sulphur is used in the vulcanisation of:	rubber	steel	glass	plastic	●○○○
91. Gunpowder consists of sulphur, charcoal and:	sodium chloride	potassium nitrate	calcium carbonate	ammonium chloride	○●○○
92. Sulphur compounds used in fertilisers include ammonium sulphate and:	sulphur dioxide	hydrogen sulphide	carbon disulphide	sulphuric acid	○○○●
93. Sulphur dioxide and sulphites are used as food preservatives because they are:	oxidising agents	reducing agents	bases	catalysts	○●○○
94. SO_2 is used to preserve dried fruits and vegetables because it:	increases the pH	prevents oxidation of the food	oxidises the food	is a solid	○●○○
95. Which compound of sulphur is used to bleach wood pulp?	Hydrogen sulphide	Sulphur dioxide	Sodium sulphate	Sulphuric acid	○●○○
96. Sulphonyl chlorides are important in the synthesis of:	plastics only	fuels	sulphonamide antibiotics	dyes only	○○●○
97. Thiol compounds contain the functional group:	–OH	–SH	–COOH	$-NH_2$	○●○○
98. Sulphuric acid is a:	colourless oily liquid	yellow gas	white powder	green solid	●○○○
99. The density of concentrated sulphuric acid is about:	1.83 g cm^{-3}	1.00 g cm^{-3}	2.83 g cm^{-3}	0.83 g cm^{-3}	●○○○

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Questions	A	B	C	D	A B C D
100. Sulphuric acid is manufactured industrially by the:	Haber process	Solvay process	Bessemer process	Contact process	○○○●
101. The raw materials of the Contact process are sulphur, air and:	water	carbon	nitrogen	hydrogen	●○○○
102. The first step of the Contact process is the:	formation of oleum	dilution of acid	formation of sulphur trioxide	burning of sulphur to form sulphur dioxide	○○○●
103. Sulphur dioxide is purified in the Contact process because impurities such as arsenic oxide:	dissolve in acid	improve the yield	poison the catalyst	lower the pressure	○○●○
104. The reaction $2SO_2 + O_2 \rightleftharpoons 2SO_3$ is:	exothermic with $\Delta H = -198$ kJ	endothermic with $\Delta H = +198$ kJ	neutral	photochemical	●○○○
105. The catalyst used in the oxidation of sulphur dioxide is:	vanadium(V) oxide	platinum	iron	nickel	●○○○
106. The temperature range used in the oxidation of sulphur dioxide in the book is:	25–50 °C	900–1000 °C	450–550 °C	100–150 °C	○○●○
107. Sulphur trioxide is not dissolved directly in water because the reaction is:	impossible	very slow	endothermic	highly exothermic, causing violent boiling and splattering	○○○●
108. Sulphur trioxide is dissolved in concentrated sulphuric acid to form:	sulphurous acid	sulphur dioxide	nitric acid	oleum	○○○●
109. Oleum is also known as:	dilute sulphuric acid	fuming sulphuric acid	battery acid	sulphurous acid	○●○○
110. Sulphuric acid in water ionises to give hydrogen ions and:	hydroxide ions	sulphate ions	sulphide ions	sulphite ions	○●○○
111. Sulphuric acid reacts with a base to form salt and water. This is called:	neutralisation	hydrolysis	oxidation	dehydration	●○○○
112. Dilute sulphuric acid reacts with active metals to give a salt and:	oxygen gas	hydrogen gas	sulphur dioxide	carbon dioxide	○●○○
113. Dilute sulphuric acid reacts with sodium carbonate to give sodium sulphate, water and:	carbon dioxide	oxygen	sulphur dioxide	hydrogen	●○○○
114. Sulphuric acid acts as a dehydrating agent because it has a great affinity for:	carbon	oxygen	water	hydrogen	○○●○
115. Hot concentrated sulphuric acid acts as an:	catalyst only	reducing agent only	neutral compound	oxidising agent	○○○●
116. Hot concentrated sulphuric acid reacts with copper to liberate:	sulphur dioxide	sulphur trioxide	oxygen	hydrogen	●○○○

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Questions	A	B	C	D	A B C D
117. In the reaction $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$, the sulphur is reduced from:	0 to +4	+6 to +4	+4 to +6	+6 to 0	○●○○
118. In the same reaction, copper is oxidised from:	+2 to 0	+1 to +2	0 to +2	0 to +1	○○●○
119. Sulphuric acid is used in the manufacture of:	only glass	only plastics	only fuels	detergents, paints and dyes	○○○●
120. Sulphuric acid is used in:	candles	matches only	car batteries	bleaching powder only	○○●○
121. Sulphur dioxide is the main cause of:	photochemical smog only	ozone formation	global cooling	acid rain	○○○●
122. The oxidation of SO_2 to SO_3 in the atmosphere is catalysed by:	O_3	N_2	NO_2	CO	○○●○
123. Which statement about nitrogen fixation is correct?	It is the burning of nitrogen	It is the conversion of nitrates into nitrogen	It is the removal of ammonia	It is the conversion of nitrogen gas into nitrogen compounds	○○○●
124. The conversion of nitrates into nitrogen gas is called:	ammonification	nitrification	nitrogen fixation	denitrification	○○○●
125. The conversion of ammonia into nitrites is known as:	nitrification	denitrification	neutralisation	nitrogen fixation	●○○○
126. Ammonia solution reacts with dilute sulphuric acid to form the fertiliser:	sodium sulphate	ammonium sulphate	ammonium chloride	ammonium nitrate	○●○○
127. Ammonia reacts with nitric acid to form:	sodium nitrate	nitrogen dioxide	ammonium sulphate	ammonium nitrate	○○○●
128. A large-scale use of ammonium nitrate is as a:	fuel	fertiliser	refrigerant	dye	○●○○
129. When solid ammonium sulphate is heated with solid sodium hydroxide, which gas is evolved?	Hydrogen	Nitrogen dioxide	Oxygen	Ammonia	○○○●
130. Ammonia gas can be identified by its effect on:	moist red litmus paper (turns blue)	lead acetate paper	starch paper	moist blue litmus paper (turns red)	●○○○
131. Ammonia can be displaced from its ammonium salts by heating with a:	noble gas	base	dilute acid	neutral salt	○●○○
132. Ammonia gas is:	brown and acidic	odourless and neutral	coloured and acidic	colourless, alkaline with a pungent smell	○○○●
133. The formula of nitrogen dioxide is:	NO	N_2O	NO_2	N_2O_5	○○●○

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Questions	A	B	C	D	A B C D
134. The formula of nitrous oxide is:	NO_2	N_2O	NO	N_2O_4	○●○○
135. Water vapour is always present in car exhaust fumes because it is formed by the:	burning of nitrogen	evaporation of engine oil	reaction of CO with NO	combustion of hydrocarbon fuel	○○○●
136. Car exhaust remains harmful even with a catalytic converter because it still contains:	carbon dioxide	argon	helium	oxygen	●○○○
137. Most of the remainder of the exhaust gases of a petrol engine (besides those in the table) is:	hydrogen	argon	helium	nitrogen	○○○●
138. In the Contact process, sulphur trioxide is absorbed in concentrated sulphuric acid to form:	dilute sulphuric acid	sulphurous acid	sulphur dioxide	oleum ($H_2S_2O_7$)	○○○●
139. Oleum is diluted with water in a controlled manner to produce:	hydrogen sulphide	concentrated sulphuric acid	sulphur	sulphur dioxide	○●○○
140. Sulphur obtained from impurities in fossil fuels can be used as a raw material in the:	Ostwald process	Haber process	Solvay process	Contact process	○○○●
141. Sulphuric acid solidifies at about:	−10 °C	0 °C	100 °C	10.37 °C	○○○●
142. Sulphur and its compounds are found in:	volcanoes, organic matter and minerals	only water	only air	only in meteors	●○○○
143. Sulphur exposed to moist air may be oxidised slowly to:	sulphur trioxide only	sulphide ions	sulphuric acid	hydrogen sulphide	○○●○
144. Concentrated sulphuric acid chars sugar because it acts as a:	catalyst only	base	dehydrating agent	reducing agent	○○●○
145. Sulphuric acid is one of the:	chemicals not used in industry	most widely produced industrial chemicals	chemicals used only in laboratories	rarest chemicals	○●○○
146. Nitrosomonas are a group of specialised bacteria that convert:	ammonia into nitrites	nitrogen into ammonia	nitrates into nitrogen	sulphur into sulphates	●○○○
147. The solution of sulphur dioxide in water is:	basic	neutral	acidic	buffer	○○●○
148. Nitric oxide from car engines and carbon monoxide are both removed in the catalytic converter by:	cooling	filtration	reaction at the catalyst surface	dissolving in water	○○●○
149. In the Haber process, the ammonia produced is separated from the unreacted gases by:	absorbing it in alkali	burning it	filtering it	cooling to liquefy the ammonia	○○○●
150. The oxidation of sulphur dioxide to sulphur trioxide is the:	most important step in the production of sulphuric acid	last step of nitrification	first step of the Haber process	first step of denitrification	●○○○

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